Answer key: Lesson 6, Tens Jump, Ones Hop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.

Answer: 3 bundles, 6 singles, 36 sticks

The picture shows 2 bundles and 6 singles, which is 26. One more bundle makes 3 bundles, and the 6 singles stay. 30 and 6 is 36.

Watch for: Adding the bundle to the singles (27).; Forgetting the old bundles (16).

S2. In the box, draw 3 bundles of ten and 4 single sticks. Next, draw 2 more bundles of ten. How many sticks are there altogether?

Answer: 54 (5 bundles drawn and 4 singles)

3 bundles and 4 singles is 34. Two more bundles add 20, so there are 5 bundles and 4 singles: 54.

Watch for: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bundles of ten.

S3. Jump forward. Fill in the blanks.

Answer: 33, 63, 81, 67

(a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.

Watch for: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).

S4. Jump or hop. Fill in the blanks.

Answer: 24, 38, 55, 69

(a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.

Watch for: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).

S5. Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

Answer: 38 pencils

3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.

Watch for: Taking away 3 (65).; Adding instead of taking away (98).

S6. Siti says 45 + 20 = 47. Fill in a to c.

Answer: Wrong. The right answer is 65. She added 2 ones instead of 2 tens.

20 is 2 tens, so Siti should jump two tens: 45, 55, 65. She added 2 ones and got 47. A jump changes the tens digit.

Watch for: Saying she is right because 20 has a 2 in it.; Not saying what she mixed up (any clear sentence about tens and ones is fine).

S7. Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

Answer: 56 (two jumps of ten from 36: 46, then 56)

36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.

Watch for: Drawing 2 small hops (38).; Changing the ones digit.

Think harder (page 3)

T1. Add 34 + 25. Show your jumps and hops on the lines.

Answer: 59 (jump 20 to 54, then hop 5 to 59)

25 is 2 tens and 5 ones. 34 + 20 = 54. Then 54 + 5 = 59. Check: 3 tens and 2 tens are 5 tens; 4 ones and 5 ones are 9 ones.

Watch for: Adding only the tens and stopping at 54.; Adding 2 + 5 as 7 tens (74).

T2. Subtract 68 - 24. Show your jumps and hops on the lines.

Answer: 44 (jump back 20 to 48, then hop back 4 to 44)

24 is 2 tens and 4 ones. 68 - 20 = 48. Then 48 - 4 = 44.

Watch for: Taking 24 away from the ones only.; Hopping forward instead of back.

T3. Frog A starts on 12. It jumps 3 tens, then hops 4. Frog B starts on 12. It hops 4, then jumps 3 tens. Fill in a to c.

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Watch for: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

T4. Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Answer: 48 (jump back 3 tens from 78)

Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Watch for: Adding 30 to 78 (108).; Taking away only 3 (75).

Go further (page 4)

G1. Mei is on 7. She jumps in tens until she gets to 97. Fill in a and b.

Answer: 9 jumps: 17, 27, 37, 47, 57, 67, 77, 87, 97

7 + 10 = 17, then 27, 37, 47, 57, 67, 77, 87, 97. That is 9 jumps. The ones digit stays 7. Check: 97 is 9 tens more than 7.

Watch for: Saying 10 jumps (counting the start).; Stopping at 87.

G2. Mei goes from 7 to 97 again. This time she can jump 10 or jump 20. Fill in a and b.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order)

She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Watch for: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

G3. Start at 25. Jump forward 3 tens. Then jump back 1 ten. Then jump forward 2 tens. Then hop forward 4. Fill in a and b.

Answer: 69 (25, 55, 45, 65, 69)

25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. 65 + 4 = 69. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65, then 4 more.

Watch for: Jumping forward on the jump back.; Forgetting the hop of 4 (65).

Investigation (page 5): Jump road

From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30 or 40) can land on 50.
Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.

Exit ticket (page 5)

E1. Add.
Answer: 76
4 tens and 3 tens make 7 tens. The 6 ones stay: 76.
Watch for: Writing 49 (adding 3 to the ones).; Writing 16.

E2. Subtract.
Answer: 44 and 71
(a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.
Watch for: Doing the same for both parts.; (b) writing 44.

E3. Add.
Answer: 59
28 is 2 tens and 8 ones. 31 + 20 = 51, then 51 + 8 = 59.
Watch for: Stopping at 51.; Writing 39.

Sneak-peek puzzles (not in the workbook)

1. 36 stars.
2. No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.